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A generalized grey model with symbolic regression algorithm and its application in predicting aircraft remaining useful life

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ABSTRACT

As a sparse data analysis method, a grey model faces challenges in interpretability for its effective application in uncertain systems. This study proposes a generalized grey model (GGM) based on symbolic regression, designed to improve the intelligence and adaptability of grey models. The GGM serves as a unified framework, integrating various grey model families and addresses regression challenges to determine the model structure. Symbolic regression in the GGM identifies symbolic input-output relationships, offering an interpretable approach for structure determination. By leveraging the non-uniqueness principle in grey system theory and employing structural penalty parameters, the model balances complexity and interpretability. A comparative analysis between GGM and conventional grey function models is conducted focusing on the differences in modeling, structure identification, and parameter optimization. Validation on the M3 competition dataset demonstrated the GGM's superior performance, achieving a significant reduction in prediction error compared to other grey forecasting models. Additionally, a rigorous analysis of aircraft lifespan data underscored the robustness and accuracy of GGM in practical engineering applications.

1. Introduction

1.1. Research background and purpose

State-space equations for dynamic systems are crucial predictive tools utilized in various fields such as physical sciences, engineering, and economic management (Arnold and King, 2021). Several time series forecasting algorithms, including classical statistical regression, exponential smoothing (Ciner et al., 2024), and empirical degradation models (Kartal et al., 2024), serve as versatile predictors for different scenarios. Grey prediction models, which treat complex systems as grey boxes, use differential equations to characterize a system's state (Liu and Forrest, 2010). Fig. 1 illustrates the prediction mechanism of the grey forecasting model. By utilizing the accumulation generation operator (AGO) and the inverse accumulation generation operator (IAGO), the grey forecasting model can construct various state-space equations to achieve time series prediction. Consequently, they find frequent applications in the sparse data analysis of intricate and uncertain systems.

In recent years, the emergence of advanced data analysis algorithms, particularly those leveraging machine learning (ML), has facilitated the accurate estimation of observational dynamics and the automatic generation of mathematical expressions for constructing models (Ren et al., 2022). ML methods present distinct advantages over other approaches, offering interpretable model learning, flexibility, and the capability to integrate diverse models. This adaptability enables ML methods to be applied across a broad spectrum of domains and problem sets. Therefore, the primary aim of this study is to incorporate interpretable ML methods into grey prediction models to improve their intelligence and overall comprehensibility.

1.2. Research motivation

In order to increase the intelligence of grey prediction algorithms

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and their adaptability in complex data analysis tasks, many scholars have improved the structure of grey models (Liu et al., 2024; Wu et al., 2024; Xu et al., 2024). As shown in Fig. 2, existing grey models have various structural representations that can be applied to different forecasting tasks. However, all the improvements are based on a hypothetical model to adapt to the evolving state of the uncertain system in concern, which is often not compatible with reality (Liu et al., 2024). Therefore, the upcoming sections will focus on addressing specific challenges from the standpoint of interpretable modeling. The primary objectives of this manuscript are outlined as follows:

(1) Various grey models employ diverse basis functions to construct grey function models tailored to different scenarios, requiring significant expertise. The central challenge lies in selecting an appropriate model structure with limited feature information. Is it possible to devise state equations that do not rely on basic functions yet achieve comparable performance? Hence, the development of a dynamic and comprehensive model structure equation is imperative to bolster the adaptability of grey models.
(2) Grey models conceptualize complex systems as grey boxes, where decision-making processes and error propagation often remain obscure. As shown in Fig. 1, the modeling process of the grey models involves a crucial structure identification phase. In intricate engineering domains with sparse and fluctuating data, employing a fixed basis function model structure to simulate system states proves arduous. Model interpretability assumes paramount importance for error diagnosis, understanding flawed decisions, and subsequently optimizing the model. It is imperative to scrutinize the input-output relationships between feature sequences and behavior sequences from an intelligible perspective. Emphasizing the interpretability of the model mechanism is indispensable for mitigating unwarranted state equations in complex application environments.
(3) Given that grey models' adaptation scenarios frequently entail small-sample time series data analysis tasks, the available data volume in these scenarios is typically limited. Inadequate training samples curtail the model's generalizability for future estimation, thereby making it imperative to extract more valuable information from scant data. Consequently, adopting a flexible and adaptive approach to strike a balance between model accuracy and complexity emerges as a superior choice for addressing practical application problems.

1.3. The contributions of the study

This study used symbolic regression methods to enhance the accuracy of a grey forecasting model in small-sample time series forecasting tasks. Existing improved models (Balochian and Baloochian, 2021) still require a predetermined system assumption, limiting the adaptability of grey models. In contrast to other competing algorithms, the proposed method offers unique advantages in terms of intelligence and interpretability. The main contributions of this study are as follows:

(1) Explicit expressions for the generalized grey model (GGM) are derived based on various grey function models. The proposed GGM can represent existing grey linear models, additive models, and nonlinear function models, while also deriving novel grey differential equation structures.
(2) The GGM incorporates symbolic regression to enhance model interpretability by structurally identifying its components. This method transforms the function fitting challenge in time series analysis into a supervised learning task. Through genetic programming, symbolic regression adaptively generates tree-like model structures.
(3) Compared with existing grey modeling methods, the proposed GGM reduces the expert experience required for the use of grey models. There is no need to predefine a complex model structure, and the proposed GGM combined with symbolic regression can adaptively fit the observed data, reducing the dependence on expert knowledge in the modeling process. The Case study has also confirmed the accuracy of the proposed method.

1.4. Organization of the study

The remaining structure of this manuscript is organized as follows: Section 2 presents the literature review of time series algorithms along with the necessary preliminary knowledge. Following this, an in-depth discussion on the utilization of the elastic-net regression method for the structural optimization of grey function models is provided. In Section 3, the proposed GGM is introduced, with its characteristics and the process of deriving its explicit expression being explained. The critical issue of structural identification is addressed through the introduction of a symbolic regression method based on genetic programming. Section 4 uses several datasets to validate the performance of the proposed GGM against other existing grey models. Section 5 showcases the practical accuracy of GGM through a specific engineering application, covering the application of different models in real-world problems and an in-depth discussion of their effectiveness. Section 6 summarizes the main research findings and future research directions, aiming to stimulate further discussion on grey models and interpretable modeling.

Nomenclature

x_t	The original observations
$x_t^{(1)}$	First-order accumulation data
h_i	Time intervals of the time variable
GM(1,1)	Grey model
GGM	Generalized grey model
PTGM	Polynomial time grey model
LSTM	Long short-term memory model
SVR	Support vector regression model
Γ	Function set of symbolic regression algorithm
$T_{\max}$	Maximum generations of symbolic regression algorithm
N	The population size of symbolic regression algorithm
MAPE	Mean absolute percentage error
RMSE	Root mean square error
MAE	Mean absolute error

2. Literature review

2.1. Machine learning algorithms for time series prediction

Data-driven ML methods serve as effective mathematical tools for time series forecasting tasks. Among the frequently utilized techniques are neural network algorithms, employing multi-layer perceptrons to establish network-like structures and adaptively optimize parameters through feedback regulation (Kurani et al., 2023). Recurrent neural networks (RNNs) (Pang et al., 2020), characterized by recurrent connections, excel in capturing long-term dependencies in time series data, making them suitable for tasks requiring prolonged memory dependence, such as natural language processing and stock price forecasting. Long short-term memory networks (LSTM) (Moghar and Hamiche, 2020), a specialized form of RNN, incorporate memory units to handle long-term dependencies more effectively. For time series forecasting of continuous target variables, decision trees or regression forest algorithms also prove effective (Klusowski and Tian, 2024). These algorithms exhibit exceptional performance on large datasets, providing lower memory consumption and faster training speeds.

All the aforementioned algorithms showcase remarkable performance in time series forecasting tasks. Given that time series data inherently contain noise and measurement results are uncertain, it is imperative to quantify uncertainty using advanced intelligent models (Abbaszadeh Shahri et al., 2022b). Data-driven ML methods

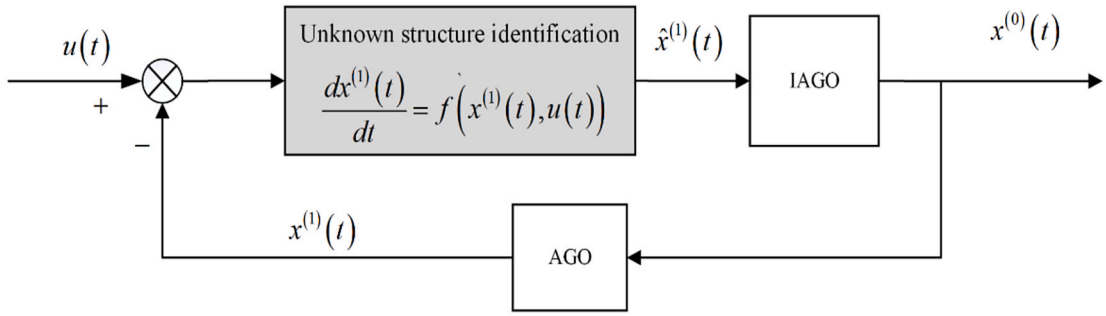


Fig. 1. Schematic diagram of state space equations for grey model.

autonomously learn patterns and relationships from data, aiding in the discovery of new knowledge and insights. However, in many real-world systems, complexity is inherent, and the interpretability of ML methods (Lu et al., 2024) has increasingly captured researchers' attention. Interpretability, crucial for diagnosing model errors and addressing biases, refers to the ability to understand and explain a model's decision-making process and its response to input data.

2.2. Symbolic regression method

In this study, the symbolic regression algorithm, a transparent and interpretable ML technique, is employed to unveil deterministic relationships within dynamic systems. In symbolic regression, symbolic derivation and algebraic expressions are used in the ML context, aiming to discern the symbolic connections between inputs and outputs. The algorithm navigates a tree-like function space, confined by pre-selected mathematical operators and operands (variables, constants, etc.), with a population of randomly generated candidate solutions being utilized (Zojaji et al., 2022).

The tree-like function structure serves as a foundational function approximator, continuously evaluated and optimized to align with observed outputs through programming. Recently, neural networks have been integrated with symbolic computation, enhancing the model's reasoning and interpretative abilities. This fusion expands the flexibility of symbolic regression, enabling it to tackle more intricate challenges. Additionally, combining symbolic regression with reinforcement learning has proven effective for managing real-world uncertainties, particularly valuable for addressing inherent uncertainties in practical scenarios (Beyhan, 2021).

2.3. Expansions and limitations of grey models

Grey models employ differential equations to describe the uncertainty relationship between the increment and cumulative quantities of a system, making them a reliable method for time series forecasting (Ding et al., 2024; Balochian and Baloochian, 2021) and multi-criteria decision-making processes (Radovanović et al., 2024; Phulara et al., 2024). There are two main types of grey models: continuous-time grey models using ordinary differential equations and discrete-time grey models using generalized difference equations. Both types exhibit various model structures adaptable to linear and nonlinear prediction tasks. The fundamental form of a grey model is abbreviated as GM(1,1), where the first "1″ represents first-order differentiation, and the second "1″ represents one variable. Scholars have enhanced the structure and computation of traditional grey models, forming a comprehensive framework with several key aspects:

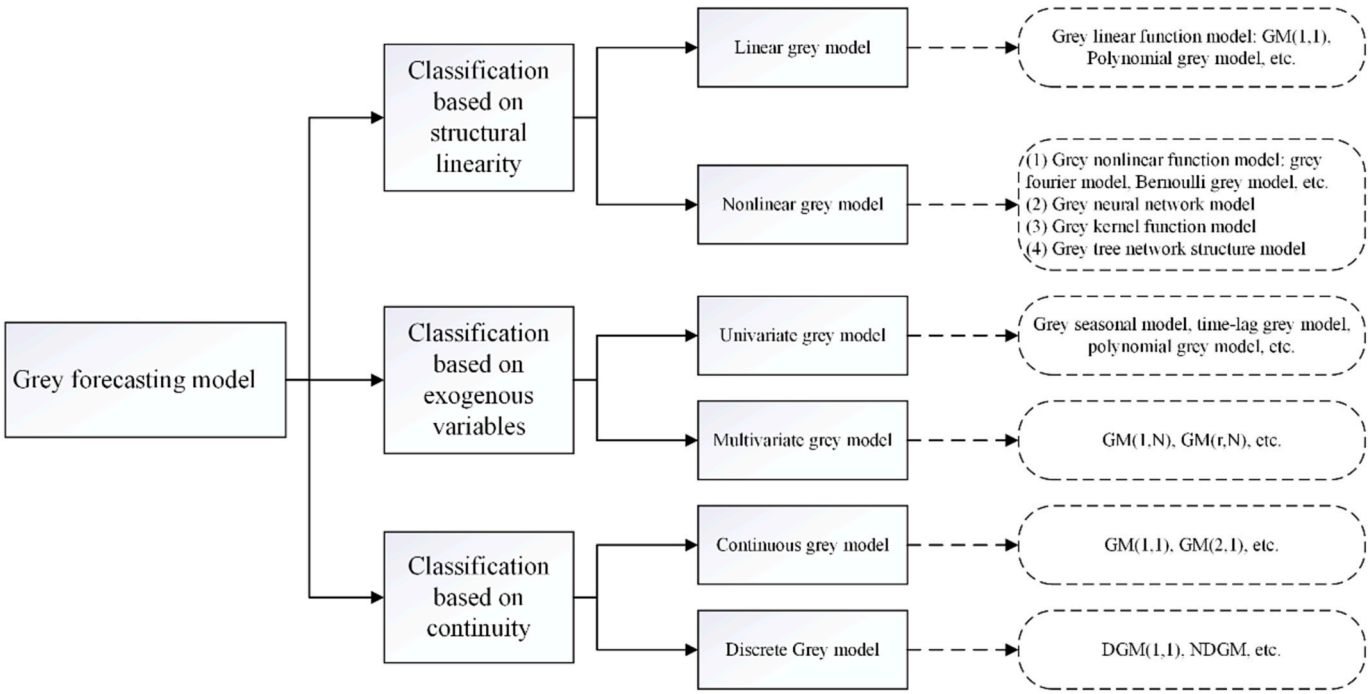


Fig. 2. Classification of existing grey models.

(1) Grey generation operators:

Grey generation operators constitute a unique data preprocessing method in the theory of grey systems (Wang and Zhang, 2022). The traditional first-order accumulative grey generation operator uses grey generation operations to generate first-order accumulative values, reducing the smoothness of observed data. To uncover hidden features in the data, Wu et al. (2013) proposed a fractional accumulative operator as an optimized form of first-order accumulation. Building upon this, Ma et al. (2020) introduced the Conformable fractional accumulative operator, reducing the complexity of fractional calculations and gaining widespread attention from scholars. A damping accumulative grey generation operator (Liu et al., 2021) was proposed to adjust the growth trend of data for more reasonable forecasting results.

(2) Parameter optimization methods:

Traditional grey models use ordinary least squares methods to solve structural parameters, which is the simplest and most reasonable approach for linear models. However, if the model includes nonlinear structural terms, identifying nonlinear parameters becomes a crucial step in achieving optimal predictions. Establishing a cost function for the prediction task and using heuristic algorithms to solve it can save significant computation time (Ding et al., 2024). A stable parameter estimation strategy ensures the accuracy of dynamic system analysis. Weighted least squares (Xie and Liu, 2023) and recursive least squares (Liu et al., 2023) methods are utilized to enhance the computational process of grey models, considering the timely value of data information.

(3) Improved model structures:

Grey models treat uncertain systems as grey boxes, employing various grey differential equations or discrete equation structures to approximate the behavior sequences of systems (Zhang et al., 2022). Additional modeling factors are introduced into GM(1,1) to construct grey function models. Common methods to optimize structural parameters include introducing time-related terms (Zeng et al., 2023). To adapt to different application scenarios, features such as nonlinear power terms (Nie and Duan, 2023), time-delay terms (Zhou et al., 2023), and seasonal factors (Dang et al., 2023) are considered as relevant factors in the modeling structure. Furthermore, polynomial structure (Wang et al., 2022), kernel function structure, Markov function structure (Gu et al., 2023), and neural network structure (Xie et al., 2023) are proposed as improved forms of grey model structures.

Based on the paradigm of GM(1,1), researchers have developed several enhanced grey model variants to cater to diverse prediction scenarios (Liu et al., 2020; Chen et al., 2008; Ma et al., 2017; Xie et al., 2020). A concise summary of these developments is provided in Table 1. The preceding literature review provided a comprehensive overview of the research framework of grey models from a broad perspective. The primary contributions in these models are: 1) This optimization involves extending the scope of grey models from linear to nonlinear predictions, enhancing their applicability and accuracy in various complex scenarios. 2) These function models are enriched by integrating polynomial functions, trigonometric functions, or kernel functions as foundational elements, thereby offering more sophisticated and adaptable modeling options.

However, the use of existing grey models in Table 1 also has certain limitations. Firstly, selecting the appropriate model structure based on scatter plots of observational data requires specialized expertise and cannot guarantee its effectiveness. Secondly, parameter estimation in existing grey models involves a two-step process, utilizing heuristic algorithms to solve hyperparameters and employing least squares regression to estimate structural parameters. For example, the polynomial time grey model (PTGM) (Liu et al., 2020) stands out as a widely adopted grey function model, utilizing polynomial functions to effectively capture and represent variations in observed values over time. Subsequently, the elastic-net regression method (Tay et al., 2023) is commonly employed to determine the model structure of PTGM. The determination of model structure and parameters is two independent processes, adding complexity to computations. Therefore, it is necessary to explore the rationality of modeling assumptions and the effectiveness of parameter estimation methods. This study proposes a generalized grey model that eliminates the need for modeling assumptions, enabling the simultaneous estimation and optimization of both model structure and parameters.

Table 1
Comparison of optimization methods for grey models.

Model name	Modeling assumptions	Variable model structure?	Hyperparametric optimization methods
Nonhomogeneous grey model (Zeng et al., 2023)	$\frac{d}{dt}x_t^{(1)} = ax_t^{(1)} + bt + c$	No	No hyperparameters to optimize
Polynomial time grey model (Liu et al., 2020)	$\frac{d}{dt}x_t^{(1)} = ax_t^{(1)} + \sum_{i=\alpha-[\alpha]}^{\alpha} b_{[i]}t^i + c$	Yes	Heuristic algorithm
Grey power model (Chen et al., 2008)	$\frac{d}{dt}x_t^{(1)} = ax_t^{(1)} + b\left(x_t^{(1)}\right)^{\alpha}$	Yes	Heuristic algorithm
Nonlinear grey model with kernel function (Ma et al., 2017)	$\frac{d}{dt}x_t^{(1)} = ax_t^{(1)} + \sum_{i=1}^{N} b_i\phi_i(t) + c$	Yes	Regularization methods
Continuous fractional grey model (Xie et al., 2020)	$\frac{d^r}{dt^r}x_t^{(\alpha)} = ax_t^{(\alpha)} + b$	No	Heuristic algorithm

3. Methodology

3.1. Traditional grey model

The grey model employs accumulated values for differential equation modeling, thereby reducing the need for extensive assumptions and expert experience. This approach is particularly effective for short-term estimation in uncertain systems. Using the GM(1,1) model (Liu and Forrest, 2010) as an example, the modeling process of a grey model involves the following steps:

Step 1. Data preprocessing by grey generative operator.

Let $X = [x_{t_1}, x_{t_2}, \cdots, x_{t_n}]^T$ be the 1-dimension time series observations, the 1-order cumulative grey generation operator can be defined as $\alpha^{(1)}$, then 1-order accumulation generation sequence $X^{(1)} = \left[x_{t_1}^{(1)}, x_{t_2}^{(1)}, \cdots, x_{t_n}^{(1)}\right]^T$ can be calculated as follow:

$$X^{(1)} = \alpha^{(1)}X \tag{1}$$

Where,

$$\alpha^{(1)} = \begin{bmatrix} h_1 & 0 & \cdots & 0 \\ h_1 & h_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ h_1 & h_2 & \cdots & h_n \end{bmatrix}, h_i = t_i - t_{i-1}, i > 2 \tag{2}$$

To eliminate the unreasonable scaling of the initial time, it is suggested to normalize the intervals of the time variable t_i by taking the mean and normalizing the first entry, that is

$$h_i = \begin{cases} 1 & ,i=1 \\ \dfrac{t_i - t_{i-1}}{(t_n - t_1)/(n-1)} & ,i>1 \end{cases}, t_k' = \begin{cases} 1 & ,k=1 \\ t_i' + \sum_{i=2}^{k} h_i' & ,k>1 \end{cases} \tag{3}$$

The 1-order accumulation operation can be viewed as the numerical discretization representation of the integral operator based on the Euler formula (Wang et al., 2024). Hence, the physical meaning of the 1-order accumulated value can be expressed using a piecewise linear integration formula:

$$x_{t_k}^{(1)} = x_{t_1} + \int_{t_1}^{t_k} x_s ds = \begin{cases} x_{t_1} & ,k=1 \\ x_{t_1} + \sum_{i=2}^{k} h_i x_{t_i} & ,k>1 \end{cases} \tag{4}$$

Step 2. Establishing state space equations with accumulated values.

The traditional GM(1,1) method utilizes the accumulated values from equation (4) to establish a differential equation model for the original time series. This can yield the following state-space equation and the corresponding discrete observation equation:

$$\text{State space equation}: \frac{d}{dt}x_t^{(1)} = a x_t^{(1)} + b \tag{5}$$

$$\text{Discrete observation equation}: x_{t_k} = a\frac{x_{t_k}^{(1)} + x_{t_{k-1}}^{(1)}}{2} + b \tag{6}$$

Where, a and b are unknown structural parameters.

Step 3. Parameter identification of models using linear regression.

Bring observation values into explicit linear regression equation (6), one has the matrix calculation form

$$\mathrm{Y} = \mathrm{B}P \tag{7}$$

Where,

$$\mathrm{Y} = \begin{bmatrix} x_{t_2} \\ x_{t_3} \\ \vdots \\ x_{t_n} \end{bmatrix}, \mathrm{B} = \begin{bmatrix} \left(x_{t_2}^{(1)} + x_{t_1}^{(1)}\right)/2 & 1 \\ \left(x_{t_3}^{(1)} + x_{t_2}^{(1)}\right)/2 & 1 \\ \vdots & \vdots \\ \left(x_{t_n}^{(1)} + x_{t_{n-1}}^{(1)}\right)/2 & 1 \end{bmatrix}, \mathrm{P} = \begin{bmatrix} a \\ b \end{bmatrix}$$

Then, the least squares estimation of structural parameters can be obtained

$$\widehat{\mathrm{P}} = \left(\mathrm{B}^T\mathrm{B}\right)^{-1}\mathrm{B}^T\mathrm{Y} \tag{8}$$

Step 4. Multi-step ahead prediction.

Introducing structural parameters $\hat{a}$ and $\hat{b}$ into the general solution of differential equation (5), one has

$$\hat{x}_t^{(1)} = ce^{\hat{a}t} - \frac{\hat{b}}{\hat{a}} \tag{9}$$

Where, c is an undetermined constant. Set $\hat{x}_{t_1}^{(1)} = x_{t_1}^{(1)} = x_{t_1}$ as initial value, this yield $c = \frac{x_{t_1} + b/a}{e^a}$.

By employing the inverse operation $X = \alpha^{(-1)}X^{(1)}$ of 1-order accumulation, one have the time response equation of the observations

$$\hat{x}_{t_k} = \begin{cases} x_{t_1} & ,k=1 \\ \dfrac{\hat{x}_{t_k}^{(1)} - \hat{x}_{t_{k-1}}^{(1)}}{h_k} & ,k>1 \end{cases} \tag{10}$$

3.2. The proposed improved generalized grey model

Different forms of grey models increase the complexity of their utilization, especially when dealing with non-linear data features. Researchers tend to use complex model structures to enhance the accuracy of simulations, but this can reduce the model's generalizability. This study introduces a novel grey model that covers the structures of various models and uses interpretable machine learning techniques to identify the model structure.

Definition 1. With the 1-order grey generation accumulation values, the generalized grey model (GGM) can be represented as

$$g\left(x_t^{(1)}\right) = f\left(t, x_t^{(1)}\right) \tag{11}$$

Where, $g\left(x_t^{(1)}\right)$ is the target value, t and $x_t^{(1)}$ are the input feature values, $f(\cdot)$ is an unknown function.

In contrast to the existing grey prediction models listed in Table 1, the GGM does not rely on any assumed functional structure. By adopting a data-driven approach, a symbolic regression algorithm is used to achieve adaptive structure and parameter optimization for the GGM model. The non-uniqueness principle of solutions, a key tenet of grey system theory, states that there can be multiple valid solutions for a given problem (Liu and Forrest, 2010). This principle is reflected in GGM, where the mathematical expression $f(\cdot)$ is inherently non-unique, leading to uncertainty in the solutions. This characteristic aligns with the core modeling assumption of grey system theory. The relationship between GGM and existing grey model families can be summarized as follows:

Remark 1. For univariate time series, GGM encompasses both first-order and higher-order grey model families.

(1) If $g\left(x_t^{(1)}\right) = \frac{dx_t^{(1)}}{dt}$ represents the first-order difference operation, then GGM generates the GM(1,1) family, which includes all the models in Table 1.

(2) If $g\left(x_t^{(1)}\right) = a_i\alpha^{(i)}x_t^{(1)} = a_1\frac{dx_t^{(1)}}{dt} + a_2\frac{d^2x_t^{(1)}}{dt^2} + \cdots + a_p\frac{d^px_t^{(1)}}{dt^p}$ represents the higher-order difference operation, then GGM yields the GM (p,1) family, encompassing models like GM(2,1), GM(3,1), and their variants.

Remark 2. For univariate time series, GGM encompasses both linear and nonlinear grey model families.

(1) If $f\left(t, x_t^{(1)}\right) = bx_t^{(1)} + ct + d$, GGM degenerates into the traditional linear grey model (Zeng et al., 2023)

(2) If $f\left(t, x_t^{(1)}\right) = bf_3\left(x_t^{(1)}\right)\cdot f_4(t) + cf_5(t) + d$, GGM yields the nonlinear grey model family, such as the non-homogeneous time grey model (Zeng et al., 2023), grey power model (Liu et al., 2022), polynomial grey model (Liu et al., 2020), grey kernel function model (Ma et al., 2017), and others.

Remark 1 and Remark 2 highlight that GGM not only encompasses existing first-order or higher-order difference grey models but also facilitates the derivation of novel forms of nonlinear grey models. While this increased complexity improves the fitting accuracy of grey models, it also presents a significant challenge in model structure identification, especially from an interpretability standpoint.

As illustrated in Fig. 3, models can be categorized into three groups based on their interpretability. Grey-box methods introduce expert knowledge to construct basic functions, simplifying the complexity of uncertain system modeling. Traditional grey function models, which

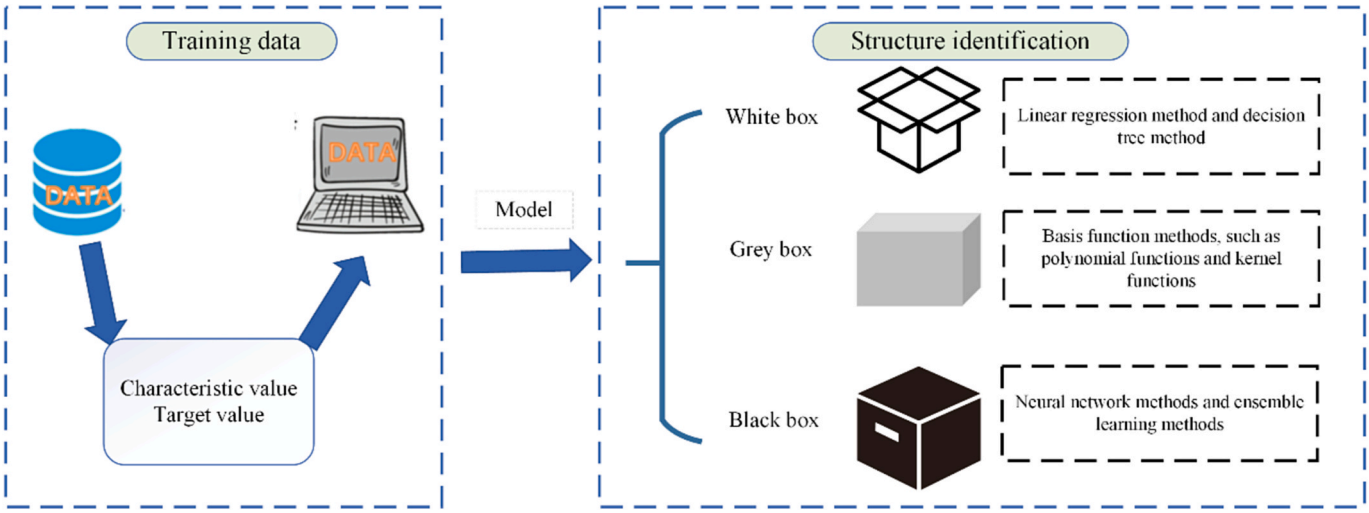


Fig. 3. Structural identification methods of generalized grey model.

employ a set of fundamental functions to simulate the system's state, fall under the grey-box category. Their interpretability is moderately superior to entirely black-box methods, which lack insight into information transmission. The black-box methods are unreliable for modeling tasks with small sample sequences, as they lack necessary physical analysis and validation. In contrast, models that rely on explicit logical mechanics and data propagation paths are classified as "white-box" models. However, in grey prediction models, no attempts have been made to interpret the system modeling mechanism using white-box methods. Interpretable modeling provides a clear understanding of how models make predictions or decisions, empowering users to better comprehend the internal logic and reasoning process of the model, thereby enhancing trust and acceptance. To improve the interpretability of grey models, it is crucial to incorporate symbolic regression methods for the adaptive structural identification of GGM.

3.3. Symbolic regression based on genetic programming

Interpretability is an inherent property of models. Symbolic regression, a supervised learning method, utilizes feature variables and target variables to unveil unknown forms of symbolic expressions. Functioning as a white-box model, it seamlessly combines precision and interpretability, ensuring that the resultant model can elucidate the underlying structure and interactions of the data in a manner comprehensible to humans. Symbolic regression finds broad applications in areas such as feature selection, time series prediction, and system modeling.

The genetic programming (GP) algorithm offers several advantages over gene expression programming (GEP). Firstly, GP demonstrates superior flexibility in evolving tree-like structures, allowing for dynamic adaptation to complex and varied data patterns, essential for accurately capturing underlying relationships. Secondly, GP provides more intuitive and interpretable mathematical representations, enhancing the comprehensibility and transparency of the resulting models—an important consideration for GGM. Lastly, GP's adaptability enables it to effectively address a wide range of scenarios without reliance on predefined system assumptions, thereby enhancing the model's robustness and applicability.

Presently, there are three primary approaches to addressing symbolic regression problems: genetic programming (Zojaji et al., 2022), models based on deep neural networks (Kim et al., 2020), and models based on reinforcement learning (Diveev and Shmalko, 2021). This study employs symbolic regression in the learning of dynamic equations for modeling complex systems, leveraging known information to formulate grey differential equation models that describe uncertain systems. Given the paramount importance of data sample availability, genetic programming is selected as the method to construct the model.

Symbolic regression, unlike linear regression which is limited to linear relationships, can express a wider range of complex nonlinear relationships (including +, -, × , ÷, sin, cos, log, etc.). However, due to its expansive search space and significant computational demands, finding the optimal solution using symbolic regression algorithms can be time-consuming. Furthermore, the outcomes are influenced by factors like the initial population and the algorithm parameters. As a result, careful selection and tuning of these parameters, as well as thorough validation, are crucial for effective symbolic regression. The method for searching the optimal formula using symbolic regression is outlined in the following steps:

Step 1. Initialize by setting the arithmetic symbol set Γ and determining runtime parameters, including termination conditions.
Step 2. Generate the initial population G_0 by randomization method.
Step 3. Assess the initial population, evaluating based on the discrepancy between predicted and actual data results and the complexity of the formulas.
Step 4. Create a new population G_t from the existing one through processes such as duplication, crossover, and mutation (refer to Fig. 4 for details).
Step 5. Iterate steps 3 and 4 until the termination criteria are satisfied, then rank the populations based on the results of these evaluations.

In this study, the Gplearn package (O'Neill, 2009) is employed for implementing symbolic regression. The specific parameter settings used in symbolic regression are detailed in Table 2. To evaluate the accuracy of the formulas derived, the mean absolute error (MAE) is used as a metric for assessing their conformity with the data.

$$MAE = \frac{1}{n}\sum_{t=1}^{n}\left|g\left(x_{t_i}^{(1)}\right) - \widehat{f}\left(t_i, x_{t_i}^{(1)}\right)\right| \tag{12}$$

Then, the objective function of symbolic regression can be expressed as a mixture of prediction accuracy and complexity

$$\min fitness_t(i) = MAE(G_t(i)) + P{\cdot}L(G_t(i)) \tag{13}$$

Where, $fitness_t(i)$ is the fitness value of population i, $MAE(G_t(i))$ represents the prediction accuracy of formula representation $G_t(i)$, $P > 0$ is parsimony coefficient, $L(G_t(i))$ represents the complexity of the dendrogram.

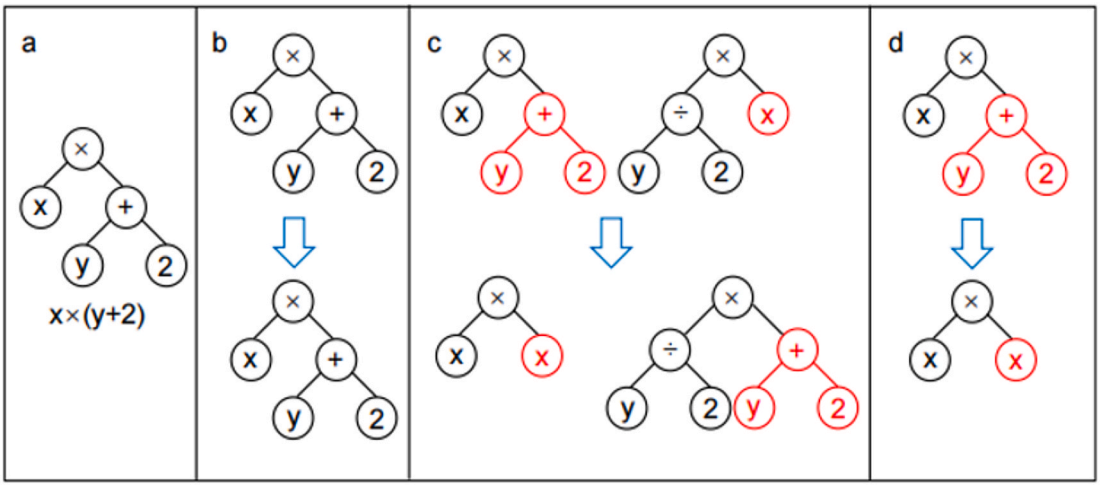


Fig. 4. Operations in symbolic regression based on genetic programming. (a) Formula representation with binary trees; (b) Copy operation; (c) Crossover operation; (d) Mutation operation.

3.4. Modeling step of GGM

This study utilizes symbolic regression to optimize the structure of GGM. Unlike methods reliant on numerical optimization and extensive data, symbolic regression is based on inference and algebraic expressions, necessitating less training data. This approach is particularly advantageous in scenarios where data is limited or costly to obtain. As illustrated in Fig. 5, the model comprises two primary steps: the first step involves representing the structure of grey differential equations, and the second step focuses on the structural identification of these equations using symbolic regression.

Typically, the time complexity of an algorithm increases with the problem size. If the population size is N, the GGM algorithm evolves for $T_{\max}$ generations, and the sample size of the observed data is n, the time complexity of the GGM algorithm is about $O(n^2 T_{\max} N)$. Where n is usually a polynomial level of the problem size, N and T depend on the details of the algorithm implementation.

Algorithm 1. Modeling steps of GGM with symbolic regression

Input: Observations $X=\left[x_{t_1},x_{t_2},\cdots,x_{t_n}\right]^T$, parsimony coefficient P, crossover coefficient C

Function set Γ, population size N, maximum generations $T_{\max}$

0: $X^{(1)}=\alpha^{(1)}X$ ◁ Generate grey accumulation sequence

1: $g\left(x_t^{(1)}\right)=f\left(t,x_t^{(1)}\right)$ ◁ Construct the model assumption of GGM

2: $G_0(i)\leftarrow random(\Gamma),\ i=1,2,\cdots,N$ ◁ Initialization

3: While $t\le T_{\max}$

4: $fitness_t(i)\leftarrow G_t(i)$ ◁ Calculate the fitness of population

5: $evaluate\left(fitness_t(i)\right)$ ◁ Evaluate and rank the performance

6: $G_{t+1}(i)\leftarrow select\left(fitness_t(i),G_t(i),P\right)$ ◁ Selection of genetic operators

7: $t\leftarrow t+1$

8: End while

Out: Predictions $\hat{X}$ and optimal formula G_{best}

Algorithm 1 gives the specific steps of the proposed GGM combined with symbolic regression. To achieve model adaptability, the parsimony coefficient and crossover coefficient are crucial parameters. These parameters play a significant role in controlling the complexity and convergence of the GGM model, ensuring that it remains both efficient and accurate while adapting to various data patterns. The parsimony coefficient helps prevent overfitting by penalizing overly complex models, while the crossover coefficient facilitates the combination of different model structures to explore a diverse solution space effectively.

Table 2
Parameters setting for symbolic regression based on genetic programming.

Term	Description
Object	Find program $f(\cdot)$ whose output matches $g\left(x_t^{(1)}\right)$ with input $\left\{t, x_t^{(1)}\right\}$
Function set:	$\{+, -, \times, \div, \sin, \cos, \log\}$
Constant range	(-1000,1000)
Fitness	MAE
Initial pop	ramped half-and-half (depth 1 to 2. 50% of terminals are constants)
Default parameters	50% subtree crossover, 25% reproduction, 25% subtree mutation
Parsimony coefficient (penalize program complexity)	0.04
crossover coefficient	0.7
Population size	2000
generations	50
Termination	Individual with fitness better than 0.1 found

Then, the structural identification problem of GGM is transformed into a supervised learning regression task. For a first-order differential equation, equation (11) degenerates into

$$\frac{dx_t^{(1)}}{dt} = f\left(t, x_t^{(1)}\right) \tag{14}$$

Using backward differencing to separate and discretize first-order difference terms, one has

$$x_{t_i} = f\left(t_i, x_{t_i}^{(1)}\right), i = 1, 2, \cdots, n \tag{15}$$

Import the explicit regression expression of equation (15) into the symbolic regression algorithm. The dataset $\left\{\left(t_i, x_{t_i}^{(1)}\right)\right\}_{i=1,2,\cdots,n}$ is used as input data, true observations $\{x_{t_i}\}_{i=1,2,\cdots,n}$ are used as the corresponding output label data.

For higher-order differential equation, equation (11) can be defined as

$$a_1\frac{dx_t^{(1)}}{dt} + a_2\frac{d^2x_t^{(1)}}{dt^2} + \cdots + a_p\frac{d^px_t^{(1)}}{dt^p} = f\left(t, x_t^{(1)}\right) \tag{16}$$

By utilizing the finite difference algorithm to approximate the differential terms, it can be simplified to

$$x_{t_i}^{(-1)} = \frac{d^2x_{t_i}^{(1)}}{dt_i^2} = \frac{\frac{dx_{t_i}^{(1)}}{dt_i} - \frac{dx_{t_i}^{(1)}}{dt_i}}{dt_i} = \frac{\left(x_{t_i}^{(1)} - x_{t_{i-1}}^{(1)}\right) - \left(x_{t_{i-1}}^{(1)} - x_{t_{i-2}}^{(1)}\right)}{\left(t_i - t_{i-1}\right)^2}$$

$$x_{t_i}^{(-2)} = \frac{d^3x_{t_i}^{(1)}}{dt_i^3} = \frac{\frac{d^2x_{t_i}^{(1)}}{dt_i^2} - \frac{d^2x_{t_i}^{(1)}}{dt_i^2}}{dt_i} = \frac{\left(x_{t_i}^{(1)} - x_{t_{i-1}}^{(1)}\right) - \left(x_{t_{i-1}}^{(1)} - x_{t_{i-2}}^{(1)}\right)}{\left(t_i - t_{i-1}\right)^2}$$

⋮

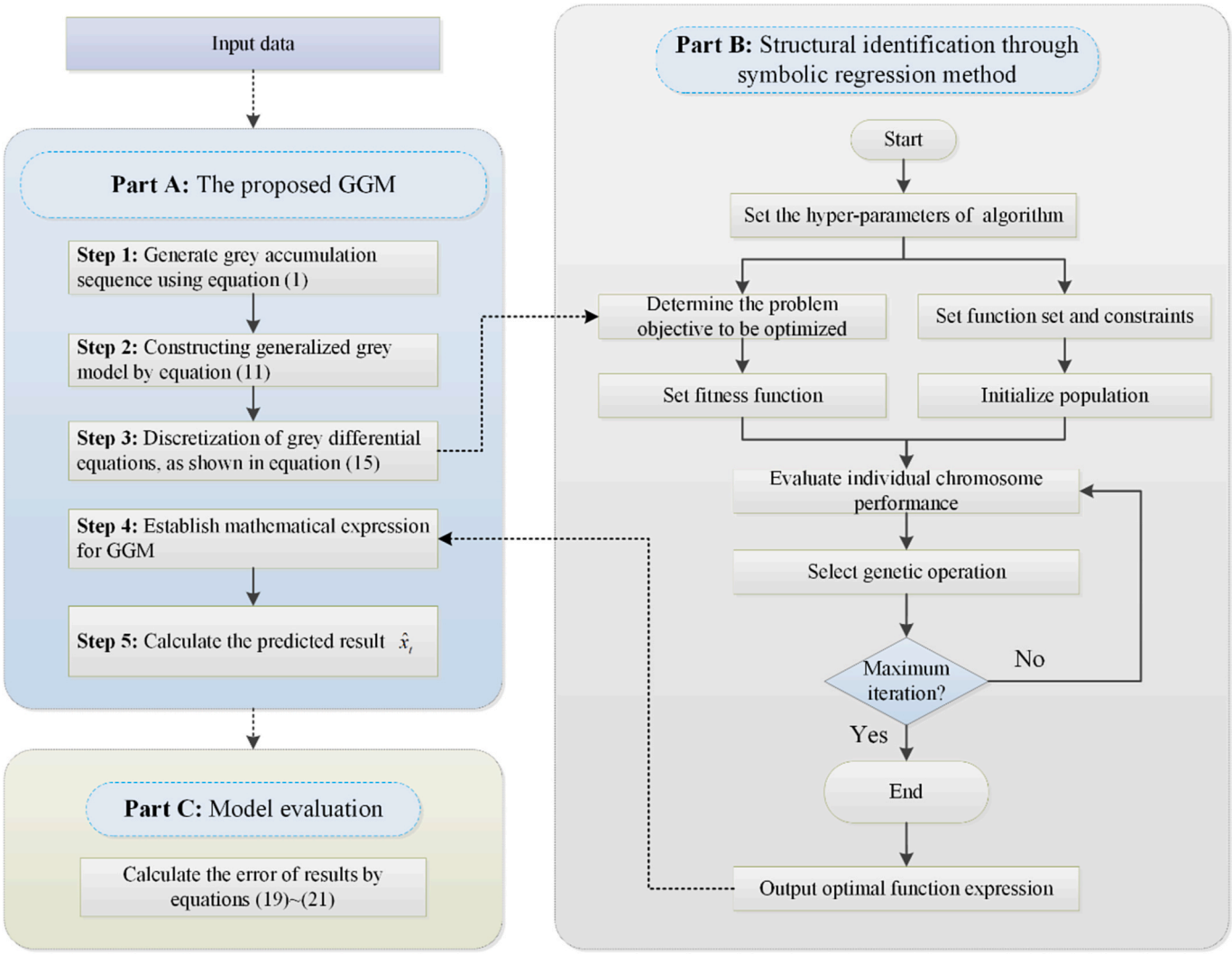


Fig. 5. Modeling steps of generalized grey model with symbolic regression.

$$x_{t_i}^{(-p+1)} = \frac{d^p x_{t_i}^{(1)}}{dt_i^p} = \frac{\frac{d^{p-1} x_{t_i}^{(1)}}{dt_i^{p-1}} - \frac{d^{p-1} x_{t_i}^{(1)}}{dt_i^{p-1}}}{dt_i}$$

Therefore, equation (16) can be approximated by the following difference equation:

$$a_1 x_t + a_2 x_{t_i}^{(-1)} + \cdots + a_p x_{t_i}^{(-p+1)} = f\left(t, x_t^{(1)}\right) \tag{17}$$

Convert it to explicit nonlinear regression

$$x_{t_i} = f\left(t, x_t^{(1)}, x_{t_i}^{(-1)}, x_{t_i}^{(-2)}, \cdots, x_{t_i}^{(-p+1)}\right), i = p, p+1, \cdots, n \tag{18}$$

Where, $\left\{\left(t_i, x_{t_i}^{(1)}, x_{t_i}^{(-1)}, x_{t_i}^{(-2)}, \cdots, x_{t_i}^{(-p+1)}\right)\right\}_{i=p,p+1,\cdots,n}$ is the input data of the GGM(p, 1) model.

As illustrated in Fig. 5, the symbolic regression algorithm takes input and output data to determine the optimal function expression. The modeling process of GGM consists of three main parts: model construction, structural identification, and model evaluation. The grey accumulated generation operator provides necessary prerequisites and computational methods for model construction. Symbolic regression algorithms are utilized for structural identification of the model, based on limited variable information. Model evaluation validates the rationality of the proposed approach and provides a basis for model improvement.

From these optimization results of symbolic regression algorithm, a clear mathematical expression for equation (11) is derived. Although higher-order differential equation (16) can be resolved using reduction methods or Laplace transforms, a first-order differential equation (14) often suffices for precise time series predictions. This study focuses solely on the structure identification and application of first-order grey differential equation models. The differential equation is solved using Python's ODE function package (Sundnes, 2023). Finally, the predicted time series values are obtained through the inverse operation of the accumulation operator, as outlined in equation (10).

4. Validation

Numerical experiments were conducted using the widely recognized benchmark dataset from the M3-Competition. The datasets were chosen to encompass various types of time series data (micro, industry, macro, etc.) and different intervals between successive observations (yearly, quarterly, monthly etc.) (Koning et al., 2005). Given the constraints of grey forecasting method, 645 samples of yearly data were selected to assess the predictive capability. Each sequence comprised more than 20 observations, with the last 6 data points reserved for testing. Descriptive statistics for the four datasets are presented in Table 3. Based on the data sample origins, the 645 time series samples were categorized into 5 groups, with varying numbers of observations per sample.

In this experiment, the proposed GGM model was compared with four existing grey models. These selected models are widely embraced in researchers for grey forecasting techniques, each possessing distinct characteristics:

Table 3
Descriptive statistics of the M3-competition datasets.

Dataset Category	Number of time series	Number of observations	Number of required forecasts
MICRO	146	20	6
INDUSTRY	102	21~47	6
MACRO	83	22~23	6
FINANCE	58	20~24	6
DEMOGRAPHIC	245	20~22	6
Other	11	22~38	6

Table 4
MAE performance of five grey models under different dataset categories.

Dataset category	GM(1,1)	FGM	NDGM	PTGM	GGM
Micro	4502.35	11824.19	56106.73	2616.58	2112.65
Industry	2026.33	1346.49	4650.06	3069.86	1730.77
Macro	494.00	600.39	645.96	587.89	1001.26
Finance	141562.07	140543.62	1081866.56	5751.29	2685.86
Demographic	1145.17	869.50	1701.52	1536.12	1561.25
Other	1299.94	1244.25	1239.71	1225.23	1373.25
Average MAE of all datasets	14589.92	15956.20	111470.19	2274.95	1738.73

(1) GM(1,1) and NDGM (Liu and Forrest, 2010) represent the classical forms of continuous and discrete grey models, respectively. They offer the advantage of not requiring additional hyperparameters and can yield results in minimal computation time.
(2) FGM (Wu et al., 2013) is a popular grey forecasting technique that incorporates fractional order differencing information from observed data into the prediction process. FGM features a hyperparameter that can be optimized through heuristic methods. It performs exceptionally well, particularly with limited data samples, adhering to the "new information priority" principle of grey system theory.
(3) PTGM (Liu et al., 2020) is a novel grey function model briefly introduced in Section 2.3. It departs from the single structure assumption of traditional models, enhancing the fitting capability of grey models. However, PTGM is prone to overfitting, especially in multi-step prediction experiments. PTGM necessitates a two-step optimization process involving heuristic algorithm-based hyperparameter estimation and elastic-net regression method-based structural parameter optimization, resulting in the highest time complexity among the five models.

Experimental results are presented in Table 4, with Mean Absolute Error (MAE) used as the error evaluation criterion. A detailed analysis of the results reveals:

(1) The proposed GGM exhibits the lowest average MAE error across all samples, indicating its superiority over competing algorithms in small-sample prediction tasks. GGM outperforms other models in the 'Micro' and 'Industry' datasets, with slightly inferior performance in the 'Macro' and 'Demographic' datasets.
(2) NDGM performs the worst across all datasets. Its characteristic amplification of temporal factors in prediction results in further inaccuracies.
(3) FGM, an enhancement of traditional GM(1,1), achieves the best performance in the 'Industry' and 'Demographic' datasets. However, its average predictive performance across all samples is weaker than GM(1,1). This is attributed to FGM's dependency on the rationality of hyperparameter estimation. Limited data samples hinder the accurate selection of suitable hyperparameters, resulting in the model's sensitivity to noise disturbances in recent data. Therefore, cross-validation methods are more reasonable for enhancing algorithm stability and reproducibility.
(4) Data-driven approaches facilitate the intelligence and generalization ability of grey forecasting algorithms. PTGM offers flexible model structures for uncertain system modeling, but the increased complexity does not significantly improve performance. Firstly, optimizing model parameters becomes more challenging with larger observation counts, making algorithm convergence more difficult. Secondly, the limited input information during model training does not support the optimization of complex model structures. GGM achieves a 23% reduction in

prediction error relative to PTGM, striking a balance between model complexity and interpretability.

5. Applications

5.1. Data source

This study utilizes monitoring data from the aerodynamic performance parameters of an aviation aircraft's airway, which are essential in aeroengine condition monitoring technology. Monitoring the airway state of an aeroengine involves two key aspects. Firstly, it involves analyzing engine performance parameter trends during steady-state cruising to aid in engine fault diagnosis (Zhao and Cai, 2024). Secondly, it includes calculating the exhaust gas temperature margin (ETGM) during engine takeoff to assess the condition of the aeroengine's hot components to monitor the engine's overall performance.

During takeoff, aeroengines typically operate at high power, leading to peak exhaust temperatures. Monitoring the exhaust gas temperature at takeoff is crucial for assessing engine health. The ETGM is commonly used as an indicator of aeroengine performance (Zhou and Huang, 2024). As an important performance parameter, ETGM can provide a visual assessment of the engine's health status. By monitoring the exhaust gas temperature margin during engine takeoff, potential issues such as degradation of high-temperature components, abnormal combustion, or turbine stall can be promptly identified and diagnosed. Therefore, ETGM not only helps airlines enhance flight safety and reduce maintenance costs but also effectively prolongs the engine's service life. Research on ETGM is of significant importance in the aviation industry as it contributes to further improving aircraft engine monitoring technology and enhancing the reliability and performance of aircraft engines. In this research, data on ETGM during a specific aircraft's takeoff mission were collected. As shown in Fig. 6, the observed data exhibit a downward trend with fluctuations over 34 sampling instances. The mean of the observed data is 79.76, with a standard deviation of 3.46. After preprocessing the data with smoothing techniques, 34 evenly spaced samples were obtained. The first 30 samples served as the training data for the model, while the remaining 4 samples were used as a validation set to evaluate the performance of various models.

5.2. Comparing models and parameter settings

To comprehensively assess the efficacy of the proposed method, this study compared it against several established time series forecasting techniques, including the long short-term memory (LSTM) model (Van Houdt et al., 2020) and the support vector regression (SVR) model (Zhang and Hong, 2021; Awad et al., 2015). Additionally, GM(1,1) and PTGM were included as comparative models based on the forecasting results presented in Table 4. Each method possesses distinct characteristics, with specific parameter and algorithm settings outlined as follows:

(1) LSTM: A type of recurrent neural network architecture, LSTM is specifically designed to overcome challenges in capturing long-term dependencies within sequential data. In this study, the LSTM model iterated 10,000 times, using a sequence of four consecutive data points as input features and the data of the subsequent time point as the target. LSTM utilizes the method of range normalization to preprocess the observed data, mitigating the influence of data scales (Van Houdt et al., 2020).
(2) SVR: This machine learning method is used for regression analysis and is particularly effective in handling non-linear problems. The SVR model in this study employed a radial basis function kernel to enhance fitting performance. It similarly utilized a sliding window approach, taking four consecutive data points as features and predicting the following data point. Then, SVR use the Z-score normalization method to preprocess the observed data (Awad et al., 2015).
(3) GM(1,1) and PTGM(1,1): The modeling procedures for these methods are detailed in Section 2.3 and Section 3.1.
(4) GGM: The input and output targets for GGM are defined according to equation (15).

To minimize the influence of heterogeneous observed data on prediction outcomes, the grey prediction model utilizes the accumulation generation operator (as seen in Equation (1)) to enhance prediction accuracy. By employing this operator, the model gains flexibility in handling unstable or highly fluctuating data, thereby effectively reducing prediction errors and enhancing model robustness. Then, mean absolute error (MAE), root mean square error (RMSE), and mean absolute percentage error (MAPE) were used to compare the multi-step prediction capabilities of these models,. These metrics provide a comprehensive evaluation of the models' accuracy in forecasting time series.

$$\mathrm{MAE}_{forecast} = \frac{1}{m}\sum_{i=n+1}^{m}\left|\hat{x}_{t_i} - x_{t_i}\right| \tag{19}$$

$$\mathrm{RMSE}_{forecast} = \sqrt{\frac{1}{m}\sum_{i=n+1}^{m}\left(\hat{x}_{t_i} - x_{t_i}\right)^2} \tag{20}$$

$$\mathrm{MAPE}_{forecast} = \frac{1}{m}\sum_{i=n+1}^{m}\left|\frac{\hat{x}_{t_i} - x_{t_i}}{x_{t_i}}\right| \times 100\% \tag{21}$$

Where, x_{t_i} is the actual observation value of the test set, $\hat{x}_{t_i}$ is the estimated value, $n = 30$ represents the number of sample points in the training set, and $m = 4$ is the number of test sample points.

5.3. Experimental results

The ETGM monitoring data is utilized to train five models, with the first-order grey differential equation (14) serving as the model structure for GGM. The Parsimony coefficient of symbolic regression is set to 0.04, indicating a preference for a simpler tree-like structure. The results of symbolic regression, depicted in Fig. 7, allow for the derivation of the mathematical expression for GGM.

In Table 5, mathematical expressions for GM(1,1), PTGM, and GGM are presented to compare linear, basis function, and symbolic regression methods in simulating uncertain system states. PTGM introduces

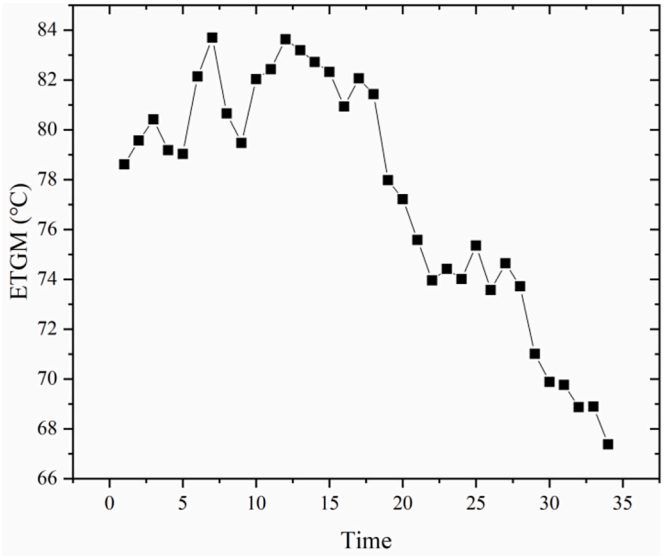


Fig. 6. ETGM observations at aircraft take-off.

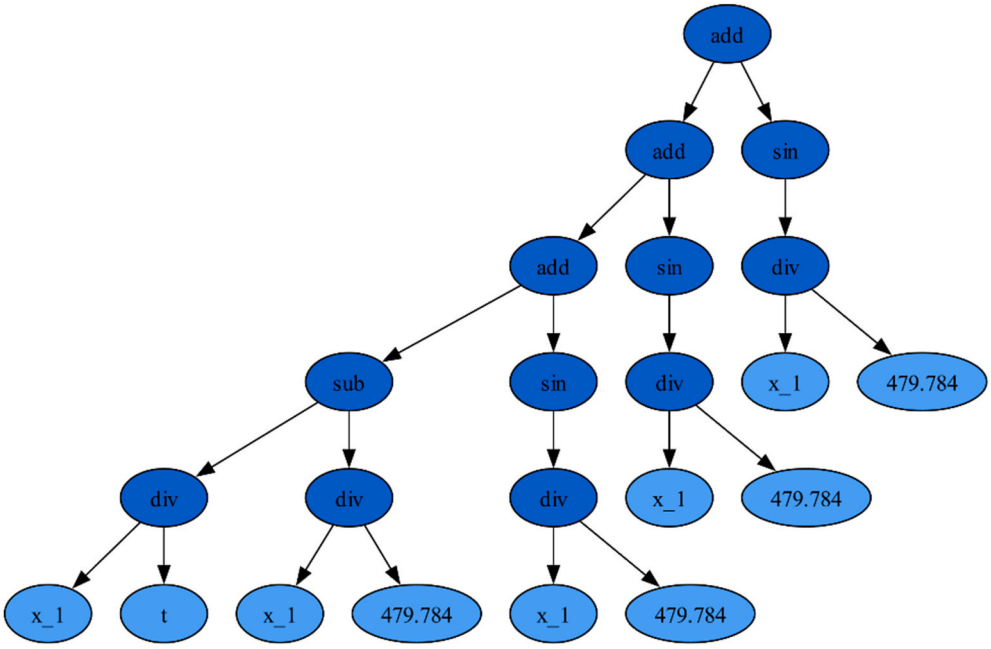


Fig. 7. Tree graph of GGM's function expression.

Table 5
Mathematical expressions for three grey models.

Models	Mathematical expressions
GM(1,1)	$\frac{dx_t^{(1)}}{dt} = 0.0046x_t^{(1)} + 84.1902$
PTGM	$\frac{dx_t^{(1)}}{dt} = 0.0212x_t^{(1)} - 1.8264t^{0.74} - 0.0056t^{2.74} + 78.7124$
GGM	$\frac{dx_t^{(1)}}{dt} = \frac{x_t^{(1)}}{t} - \frac{x_t^{(1)}}{479.784} + 3\sin\frac{x_t^{(1)}}{479.784}$

polynomial functions as a nonlinear structure, enhancing the adaptability of grey models to nonlinear systems based on traditional linear models. However, the selection and optimization of basis functions require substantial expertise. GGM, utilizing symbolic regression, introduces a more diverse function set (see Table 2), making the model structure more flexible. The tree-like structure also facilitates a visual and interpretable data-driven modeling process. Additionally, after conducting five independent trials, the GGM produced identical results each time, demonstrating the model's reproducibility. This consistent performance across multiple trials further validates the robustness of the model and reduces the necessity for additional non-parametric tests like the Wilcoxon test.

The performance of the five models is displayed in Fig. 8 and Table 6. The forecasted MAPE for LSTM, SVR, GM(1,1), PTGM, and GGM are 2.64%, 9.05%, 5.82%, 3.93%, and 1.21%, respectively. The observed data for ETGM form a fluctuating time series with a declining trend. The proposed GGM demonstrates the highest predictive performance. Fig. 8 illustrates that the prediction curve of GGM closely matches the actual curve of observed values, providing evidence of GGM's superior adaptability to complex data prediction tasks. The MAE error plot in Fig. 8 also illustrates the accuracy performance of each model in multi-step forecasting, showing that GGM is comparatively more stable than other models. The increasing prediction error of PTGM with the prediction steps indicates its inability to accurately simulate and interpret the true state of unknown systems. While the LSTM model demonstrates good performance, its mechanism is not interpretable, requiring multiple adjustments to find optimal parameters. SVR, as a statistical prediction method, does not perform satisfactorily with limited data and uncertainty. The comparative analysis yields the following insights:

(1) Traditional linear model GM(1,1) is effective in short-term forecasting with small-sample data analysis. However, the growth trend of the predicted values is solely influenced by the grey development coefficient a (see Equation (5)), limiting the applicability of the GM(1,1) model in uncertain environments.
(2) PTGM overestimates the rate of data change, relying heavily on temporal values "t," leading to an exaggeration of the observed system's changing trend and decreased prediction accuracy. The results in Table 6 also confirm that the existing PTGM model lacks intelligence and interpretability in fluctuation time series prediction, and thus the accuracy of PTGM is weaker than that of GGM and LSTM.
(3) A flexible model structure is advantageous for uncovering implicit information features in uncertain systems. GGM, with symbolic regression, does not constrain the model structure and can leverage a diverse set of basic symbols to find the optimal mathematical expression. In contrast to PTGM, GGM does not demand prior expert experience.
(4) Interpretable modeling methods, like GGM utilizing symbolic regression, facilitate understanding each feature's contribution to predictions. The tree-like structure provides an intuitive representation of relationships between variables, aiding interpretation of the decision-making process. Symbolic regression, based on reasoning and algebraic expression programming, enables the model to provide information about deeper relationships, beyond simple numerical predictions. The combination of grey prediction models and symbolic regression methods is a rational approach, especially in situations where data is scarce or costly.

Friedman test (Kaur and Kaur, 2013) is a commonly used non-parametric statistical signal test to test the accuracy differences among several modeling methods. In this study, five prediction techniques are repeated experiments on the same samples and the observations are the MAE values generated by the algorithmic predictions. The Friedman test p-value is 0.0036, indicating that there are significant non-random differences between the five prediction models. The results of the Friedman test for MAE values are shown in Table 7. It can be

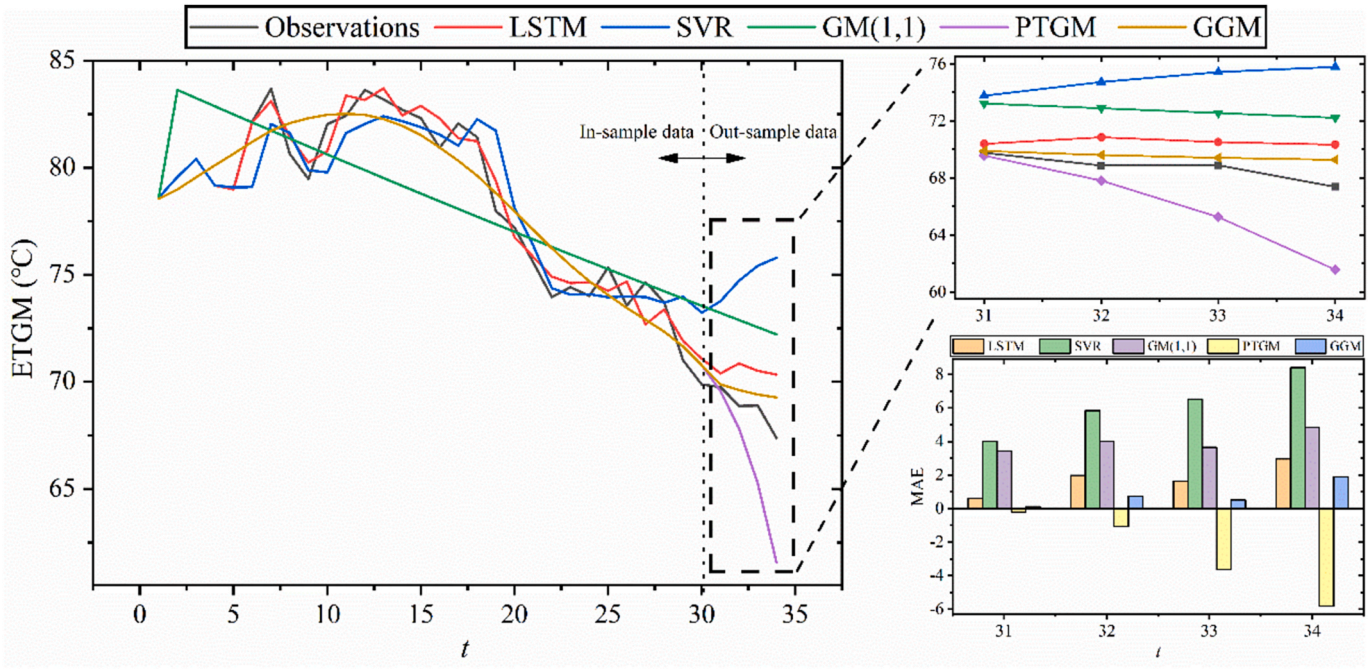


Fig. 8. ETGM prediction results of five models.

Table 6
Comparison of average prediction errors of five models.

Model	LSTM	SVR	GM(1,1)	PTGM	GGM
$MAE_{forecast}$	1.80	6.20	3.99	2.67	0.82
$RMSE_{forecast}$	1.99	6.40	4.03	3.47	1.05
$MAPE_{forecast}$ (%)	2.64	9.05	5.82	3.93	1.21

inferred that GGM is the most accurate algorithm for predicting this ETGM case. The performance of LSTM and PTGM is similar. SVR as the prediction ranked the lowest, which is also consistent with the error statistics in Table 6.

5.4. Discussion of limitations and future works

The proposed GGM model offers significant benefits, but it's important to acknowledge its limitations. Firstly, the data-driven algorithm relies on the reliability of the observed data, and fluctuations hidden in the samples may reduce the accuracy of GGM. Further consideration is needed for the impact of observation errors on parameter estimation. GGM typically requires well-designed feature engineering to enhance model performance, which may lead to suboptimal performance in simulating complex nonlinear relationships, necessitating further refinement and optimization of the algorithm.

Secondly, the mathematical expression derived from GGM is not uniquely defined. For any given problem, multiple expressions may satisfy the criteria, and the model may select one among them, introducing a degree of uncertainty in the solutions. Moreover, GGM is susceptible to sensitivity towards data noise. Its focus on symbolic relationships implies that minor data fluctuations could lead to overfitting or errors in results. To mitigate this, practical applications may necessitate techniques such as data augmentation or cross-validation to enhance GGM's adaptability to specific tasks.

Lastly, an in-depth analysis of the time correlation in the training data was not conducted, as the proposed GGM is a single-input time series prediction algorithm. However, there are alternative approaches to enhance the model's predictive performance. One such approach involves utilizing modified accumulation operators, which assign varying weights to observed values based on their temporal proximity, thus enhancing the utilization of new information (Liu et al., 2023). Another avenue for improvement is the incorporation of time delay factors into the model structure (Zhou et al., 2023).

The method presented in this study demonstrates considerable versatility for small-sample data predictions, despite not being customized for specific feature scenarios. This study points out the important application value of interpretability in time series forecasting. Several aspects require further exploration:

(1) Exploring the generalization capabilities of GGM to real-world contexts is paramount. Novice users frequently encounter difficulties in choosing an appropriate small-sample prediction algorithm from a plethora of existing grey models. Consequently, they may gravitate towards function models with elaborate structures to bolster fitting performance. Nonetheless, grey models may confront challenges in assimilating new, unobserved data, especially under conditions of limited training data or substantial shifts in the problem's paradigm. The incorporation of supplementary domain-specific insights and prior information into

Table 7
Friedman tests in M3-competition dataset for MAE values.

Modeling technique	Mean Rank
LSTM	2.25
SVR	5
GM(1,1)	3.75
PTGM	2.25
GGM	1.75

Table 8
Cross-validation test for different levels of parameters of GGM.

parsimony coefficient	crossover coefficient			
	0.8	0.7	0.6	0.5
0.02	0.57	0.51	0.51	1.85
0.04	9.88	0.82	0.51	1.27
0.06	9.88	1.74	0.51	9.88
0.1	9.88	9.88	9.88	9.88

GGM's learning step could confer benefits, particularly in situations characterized by scant data or domain expertise. Therefore, further integrating multi-output regression training models, such as the multi-output SVR algorithm (Abbaszadeh Shahri et al., 2022a; Barati and Sharifian, 2015), is a good choice, especially in situations requiring prediction of multiple related variables. By simultaneously modeling multiple output variables, multi-output SVR offers a comprehensive solution for capturing complex relationships and dependencies among various aspects of the data.

(2) It is crucial to select appropriate parameters to enhance the predictive accuracy and generalization capability of GGM. Cross-validation experiments were employed to investigate the impact of different levels of parameters on identifying the optimal parameter selection. Various levels of parsimony coefficient and crossover coefficient were utilized to generate cross-validation tables, as shown in Table 8. The parsimony coefficient represents the structural complexity of the optimization formula, while the crossover coefficient denotes the probability of crossover among the population winners during the iteration process. These two parameters respectively control the interpretability and predictive accuracy of GGM. It can be observed that the predictor's estimation results are acceptable when the parsimony coefficient ranges from 0.02 to 0.04 and the crossover coefficient ranges from 0.7 to 0.5.

(3) Striking a balance between the complexity and interpretability of the GGM model is essential, depending on the specific application requirements. Symbolic regression methods offer a practical way to modulate the model's complexity. For instance, in Section 4, adjusting the parsimony coefficient from 0.04 to 0.02 led to a more complex, tree-like model structure, as displayed in Fig. 9. The new mathematical expression for GGM is given as

$$\frac{dx_t^{(1)}}{dt} = -\frac{x_t^{(1)}}{365.458} + \frac{x_t^{(1)}}{t} + 2\sin\left(\frac{x_t^{(1)}}{730.917}\right) + \sin\left(\frac{x_t^{(1)}}{479.784}\right) + \sin\left(\frac{x_t^{(1)}}{551.143}\right) - \sin\left(\frac{x_t^{(1)}}{200.435}\right) \tag{22}$$

This adjustment improved the MAPE for ETGM data to 0.72%, slightly enhancing GGM's performance. Nevertheless, it's important to avoid overly complex structures for the sake of efficiency and computational practicality. As the "Occam's Razor" principle suggests, simpler models are often preferable due to their enhanced generalizability, computational speed, ease of interpretation, and noise resilience. Given that GGM's mathematical expression is not unique, it's advisable to opt for simpler models when two models offer similar accuracy in explaining the data.

(4) Enhancing grey models for dynamic systems improves decision-making from databases. Efficient normalization and data extraction tools can combine domain expertise with data science skills to eliminate outliers and noise (Abbaszadeh Shahri et al., 2024). The proposed GGM is designed for small sample data and offline processing. The common grey accumulated operator cannot fully handle outliers and heterogeneity in observed data. For dynamic monitoring of streaming data, integrating filtering algorithms and prediction techniques is needed for real-time updates. Future research should explore hybrid computational methods to quantify data uncertainty and automate modeling, enhancing feature extraction from limited datasets. Additionally,

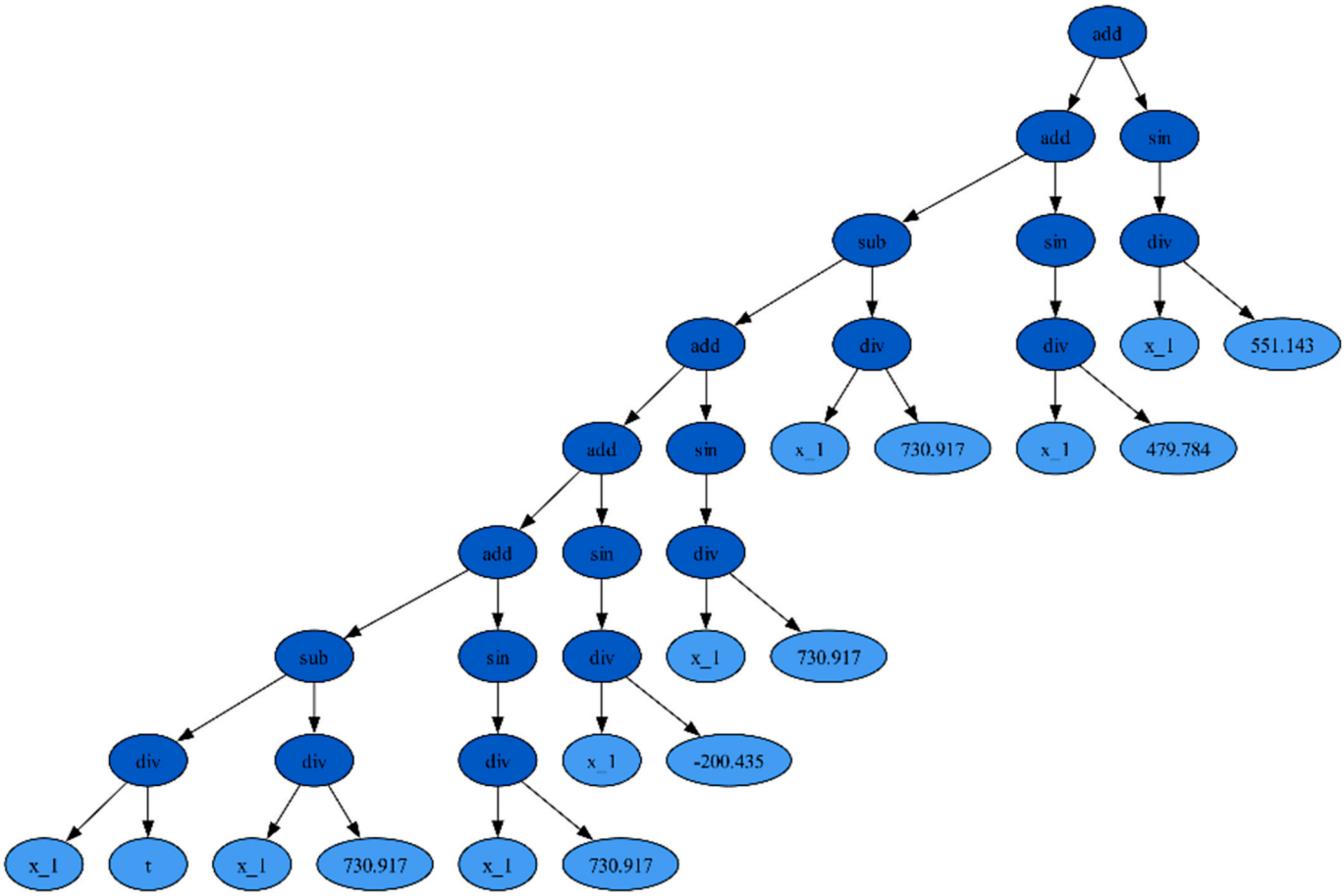


Fig. 9. Tree model structure of GGM with high complexity.

using multi-output algorithms can improve GGM performance in handling complex prediction tasks with multiple outputs.

6. Conclusions

This study introduces a novel generalized equation aimed at enhancing the adaptability of grey models for analyzing small datasets. Through a comprehensive comparison between GGM and existing grey function models in terms of modeling principles, structure, parameter identification, and prediction performance, several findings can be derived: (i) GGM represents a data-driven approach to modeling differential equations, allowing for the discovery of previously unassumed model formulations. Without requiring expert knowledge or predefined model assumptions, GGM balances model complexity and prediction accuracy by exploring a broad solution space. (ii) This study also highlights the importance of interpretability in grey model modeling and provides an intuitive view of tree model structure construction into the system's state mechanism research. Leveraging symbolic regression methods, GGM models demonstrate advantages in structure identification and parameter estimation over existing grey models. (iii) The experimental results demonstrate that the algorithm outperforms traditional grey models in both accuracy and efficiency. Additionally, in the sparse data time series forecasting task of aircraft monitoring data, GGM's performance also surpasses that of machine learning methods.

CRediT authorship contribution statement

Lianyi Liu: Writing – original draft, Methodology. **Sifeng Liu:** Funding acquisition. **Yingjie Yang:** Supervision, Resources. **Xiaojun Guo:** Validation, Software. **Jinghe Sun:** Methodology, Investigation.

Declaration of competing interest

The authors declare that they no conflict of interest.

Data availability

Data will be made available on request. The Python codes of the GGM are publicly available at https://github.com/mytruth126/GGM.

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